

Distance and Midpoint on the Coordinate Plane

Answer Key

High School Geometry

Quick Answers

- | | |
|---|--|
| 1. 10 units | 6. (a) midpoint = $(-2, -1)$, — |
| 2. $(1, 4)$ | 7. (a) length of RS = 13 units, (b) midpoint of RS = $(3, 1.5)$ |
| 3. 13 units | 8. (a) midpoint of WY = $(1, 0)$, (b) midpoint of XZ = $(1, 0)$, — |
| 4. x-coordinate of B = 7, y-coordinate of B = 8 | |
| 5. 15 | |
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What is verified

6 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 6, 8. The worked solution states what a correct response must contain.

Curriculum

Level: High School Geometry

HSG-GPE.B.4 – *problems 2, 4, 6, 7, 8*

HSG-GPE.B.7 – *problems 1, 3, 5, 7*

Difficulty 1–5 of 5 across 13 tagged checks.

Common wrong answers

- If they got 14: added the horizontal and vertical gaps instead of using the theorem*
 - If they got 10.91: subtracted the squares instead of adding them*
 - If they got 21: added the east-west and north-south gaps, which is the walking distance, not the straight line*
 - If they got 17: added the coordinate differences instead of squaring, adding and rooting*
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1. Find the distance between $A(2, 3)$ and $B(10, 9)$.

Solution. The horizontal gap is $10 - 2 = 8$ and the vertical gap is $9 - 3 = 6$; those are the legs of a right triangle whose hypotenuse is the distance.

$$d = \sqrt{8^2 + 6^2} = \sqrt{64 + 36} = \sqrt{100}$$

Therefore

$d = 10$ units

2. Find the midpoint of $C(-4, 7)$ and $D(6, 1)$.

Solution. Average each coordinate separately:

$$M = \left(\frac{-4 + 6}{2}, \frac{7 + 1}{2} \right) = \left(\frac{2}{2}, \frac{8}{2} \right)$$

Therefore

$M = (1, 4)$

3. Find the distance between $P(-5, 2)$ and $Q(7, -3)$.

Solution. Horizontal gap $7 - (-5) = 12$; vertical gap $-3 - 2 = -5$. Squaring removes the sign.

$$d = \sqrt{12^2 + (-5)^2} = \sqrt{144 + 25} = \sqrt{169}$$

Therefore

$d = 13$ units

4. $M(3, 5)$ is the midpoint of $\overline{AB}$ and $A(-1, 2)$. Find B .

Solution. The midpoint sits half way, so travelling from A to M and then the same distance again lands on B . In coordinates that is $B = 2M - A$, one coordinate at a time:

$$x_B = 2(3) - (-1) = 6 + 1, \quad y_B = 2(5) - 2 = 10 - 2$$

so

$x_B = 7$

and

$y_B = 8$

Check: the midpoint of $(-1, 2)$ and $(7, 8)$ is $\left(\frac{-1+7}{2}, \frac{2+8}{2}\right) = (3, 5)$, as given.

5. A park entrance at $(2, 1)$ and a library at $(14, 10)$, one unit per block. Find the straight-line distance in blocks.

Solution. Horizontal gap $14 - 2 = 12$ blocks, vertical gap $10 - 1 = 9$ blocks.

$$d = \sqrt{12^2 + 9^2} = \sqrt{144 + 81} = \sqrt{225}$$

Therefore

$d = 15$ blocks

6. Endpoints $E(-6, 3)$ and $F(2, -5)$. (a) Find the midpoint. (b) Explain why averaging the coordinates gives a point on the segment.

Solution (a).

$$M = \left(\frac{-6 + 2}{2}, \frac{3 + (-5)}{2} \right) = \left(\frac{-4}{2}, \frac{-2}{2} \right)$$

Therefore

$M = (-2, -1)$

Solution (b) — instructor-judged. A complete response says averaging the x -coordinates moves exactly half of the horizontal run from one endpoint, averaging the y -coordinates moves exactly half of the vertical rise, and because the segment is straight those two half-steps are taken together and land on the segment itself — half the run always pairs with

half the rise on a straight line. A response that says only "it is between the endpoints" has not explained why the point lies on the segment; award partial credit.

7. Endpoints $R(-3, -1)$ and $S(9, 4)$. (a) Find RS . (b) Find the midpoint of $\overline{RS}$.

Solution (a). Horizontal gap $9 - (-3) = 12$, vertical gap $4 - (-1) = 5$.

$$RS = \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169}$$

Therefore

$RS = 13 \text{ units}$

Solution (b).

$$M = \left(\frac{-3 + 9}{2}, \frac{-1 + 4}{2} \right) = \left(\frac{6}{2}, \frac{3}{2} \right)$$

Therefore

$M = (3, 1.5)$

8. Quadrilateral $W(-4, 1)$, $X(0, 5)$, $Y(6, -1)$, $Z(2, -5)$. (a) Midpoint of $\overline{WY}$. (b) Midpoint of $\overline{XZ}$. (c) What do the two answers tell you about the quadrilateral?

Solution (a). $\left(\frac{-4+6}{2}, \frac{1+(-1)}{2} \right) = \left(\frac{2}{2}, \frac{0}{2} \right)$, so

$(1, 0)$

Solution (b). $\left(\frac{0+2}{2}, \frac{5+(-5)}{2} \right) = \left(\frac{2}{2}, \frac{0}{2} \right)$, so

$(1, 0)$

Solution (c) — instructor-judged. A complete response observes that the two diagonals have the *same* midpoint, so each diagonal cuts the other exactly in half, and concludes that the quadrilateral is a parallelogram (a quadrilateral whose diagonals bisect each other is a parallelogram). Award full credit only when both the shared midpoint and the conclusion drawn from it appear.